

RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

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CRIME INVESTIGATION EXPERT SYSTEM

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OUTLINE

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ABSTRACT

Crime investigation is a critical process that heavily relies upon the analysis of evidence, witness statement, and suspect informations. Traditional investigation methods rely heavily on the expertise of the investigator, thus the process is time-consuming, requires more human resources, and prone to inconsistencies. This project presents a Crime Investigation Expert System that applies classic AI techniques to assist in investigations by providing intelligent, transparent, and explainable decision support.

The system utilizes structured datasets containing crime details, suspect profiles, physical evidence and etc stored in relational database. A Knowledge Base represents facts using First Order Logic and Predicate Logic, Search Algorithms (BFS/DFS), Constraint Satisfaction with Backtracking, and Forward & Backward Chaining are used to analyze crime evidence, eliminate invalid suspects, infer the prime suspect, and provide explainable reasoning for the investigation and generate a comprehensive investigation conclusion report with logical justification. . Developed using Python, Tkinter, and SQLite. The project demonstrates the practical application of classical AI concepts in crime investigation.

Keywords: Crime Investigation Expert System, Artificial Intelligence, Knowledge Representation, Search Algorithms, Constraint Satisfaction, Forward Chaining, Backward Chaining, First-Order Logic, Predicate Logic, Digital Forensics.

OBJECTIVE

- To develop an AI-based expert system for assisting in crime investigations.
- To represent crime knowledge using Predicate Logic and First Order Logic.
- To search possible suspects using Search Algorithms (DFS & BFS).
- To eliminate invalid suspects using Constraint Satisfaction and Backtracking.
- To identify prime suspect using Forward Chaining and explain them using Backward Chaining.
- To generate a comprehensive investigation conclusion report with logical justification.

Literature Survey

TITLE	YEAR	AUTHOR	TECHNIQUE
An AI Knowledge-Based System for Police Assistance in Crime Investigation	2024	Fernández-Basso et al.	Rule-Based AI, Knowledge Base, NLP
Knowledge Representation Model for Crime Analysis	2017	Mohd et al.	Ontology, Knowledge Representation
Crime Base: Towards Building a Knowledge Base for Crime Entities and Their Relationships from Online Newspapers	2019	Agarwal et al.	NLP, Rule-Based Extraction, Knowledge Base

EXISTING SYSTEM

- Crime investigations rely heavily on manual analysis by investigators.
- Evidence from different sources is analyzed separately, increasing investigation time.
- Decision-making depends largely on the investigator's experience and expertise.
- Traditional systems primarily store crime records but do not provide intelligent reasoning.
- Handling uncertain or incomplete evidence is difficult.
- There is no automated suspect ranking or explainable decision support.
- Generating investigation reports requires significant manual effort.

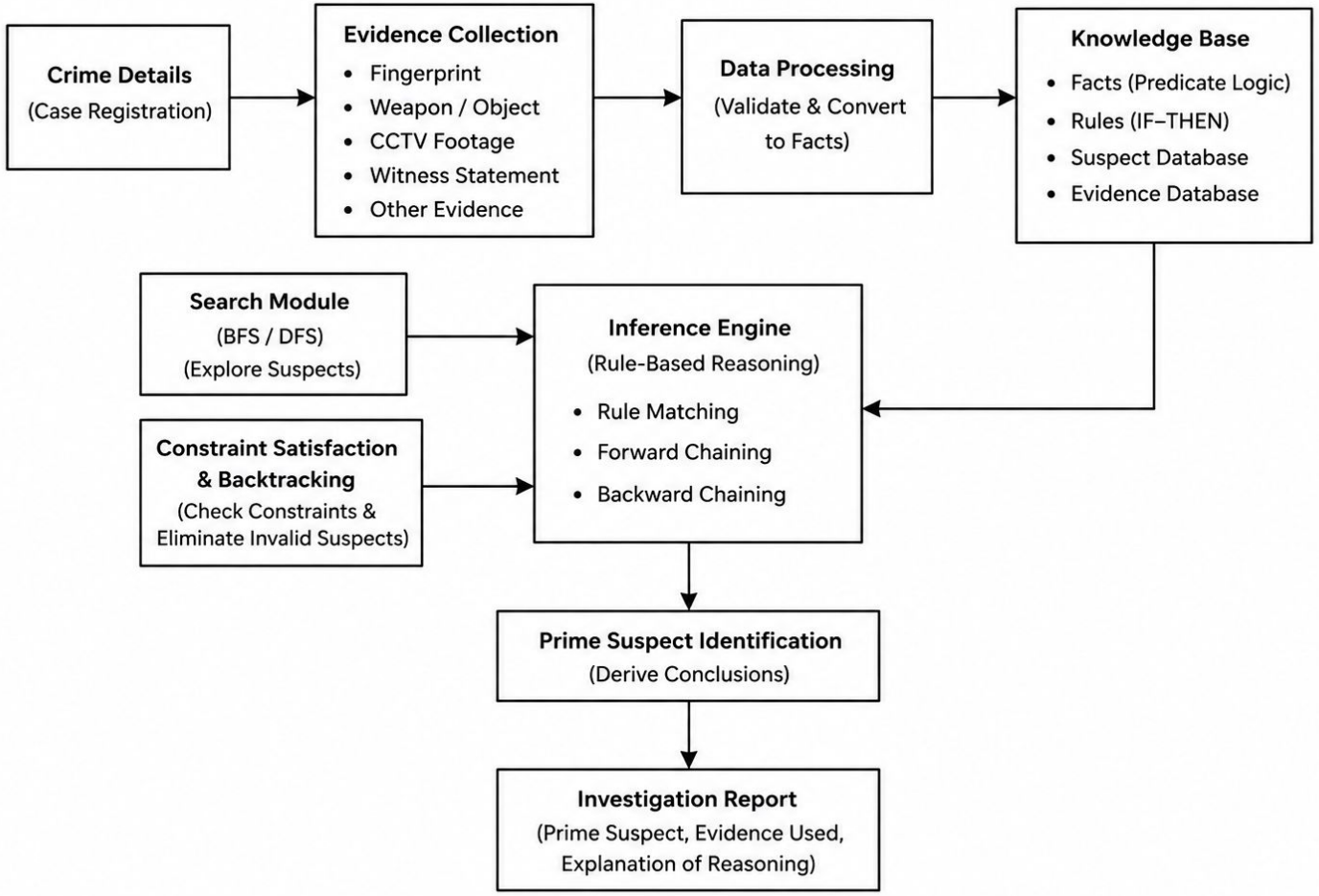
PROPOSED SYSTEM

- Utilizes structured crime datasets from Kaggle along with investigator inputs.
- Stores crime records, suspect profiles, evidence, and witness statements in an SQLite database.
- Represents investigative knowledge using Predicate Logic and First-Order Logic.
- Uses BFS/DFS to find possible suspects.
- Applies Constraint Satisfaction (CSP) and Backtracking to eliminate invalid suspects.
- Applies Forward Chaining to identify potential suspects.
- Uses Backward Chaining to explain the reasoning behind conclusions.
- Generates an Investigation Conclusion Report.

Novelty in Proposed System

- Integrates multiple Classical AI techniques from the FOAI syllabus into a single application.
- Combines BFS/DFS, Constraint Satisfaction, Backtracking, Forward Chaining, and Backward Chaining for intelligent crime investigation.
- Provides explainable decision support by tracing the logical reasoning behind every conclusion.
- Represents crime facts and investigation rules using Predicate Logic and First-Order Logic.
- Automatically eliminates invalid suspects using investigation constraints before inference.
- Generates a detailed Investigation Conclusion Report with the prime suspect, supporting evidence, reasoning, and recommendation.
- Designed as a modular rule-based expert system, allowing easy addition of new crime types, evidence, and investigation rules without changing the core logic.

BLOCK DIAGRAM(PROPOSED SYSTEM)



Hardware/Software Requirements

Hardware Requirement:

Intel Core i3 Processor or above

4 GB RAM (Minimum)

20 GB Free Storage

Windows 10/11 or Ubuntu Linux

Internet Connection (for dataset download and updates)

Software Requirement:

Python 3.11+

Visual Studio Code (IDE)

SQLite Database

Tkinter or Flask (User Interface)

Python Libraries (tkinter, sqlite3, json)

References

- **[1]** C. Fernández-Basso, K. Gutiérrez-Batista, J. Gómez-Romero, M. D. Ruiz, and M. J. Martin-Bautista, "An AI Knowledge-Based System for Police Assistance in Crime Investigation," *Expert Systems*, vol. 42, no. 2, 2024. doi: **10.1111/exsy.13524**.
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- **[3]** S. Agarwal *et al.*, "Crime Base: Towards Building a Knowledge Base for Crime Entities and Their Relationships from Online Newspapers," *Information Processing & Management*, Elsevier, vol. 56, no. 5, 2019.